REAL ANALYSIS II · SEMESTER 5 · MODULE 2

Answer key: Differentiation

Formal solutions to all 40 questions of the model paper.

BEFORE YOU READ AN ANSWER

Each answer starts with **Know first**: topic codes from File 1. If one feels new, read it first, try the question yourself, then read the solution.

Grey "Marks" notes show a typical split (a guide, not an official scheme): statement · set-up · key step · conclusion. Standard facts quoted without proof: $(x^n)' = nx^{n-1}$, $\sin' = \cos$, $\cos' = -\sin$, $(e^x)' = e^x$, $(\ln x)' = 1/x$, and $(x^r)' = rx^{r-1}$ for rational r (Plus Two facts; Bartle proves them in 6.1.10 and 8.3).

Section A · Short answers

3 marks each.

1

Definition of the derivative with ε and δ; verify $f'(2) = 4$ for $f(x) = x^2$.

KNOW FIRST

T01 The derivative: a limit of chord slopes P04 Limits of functions (Real Analysis I) M01 Limit of a difference quotient

Definition 6.1.1. Let $I \subseteq \mathbb{R}$ be an interval, $f : I \to \mathbb{R}$, $c \in I$. A real number L is the **derivative of f at c** if, given any $\varepsilon > 0$, there exists $\delta(\varepsilon) > 0$ such that if $x \in I$ and $0 < |x - c| < \delta(\varepsilon)$, then $\left| \dfrac{f(x) - f(c)}{x - c} - L \right| < \varepsilon$. We write $f'(c) = L$.

Scratch. $\dfrac{x^2 - 4}{x - 2} = x + 2$ for $x \neq 2$, and $|(x + 2) - 4| = |x - 2|$. Want this $< \varepsilon$, so $\delta = \varepsilon$ works.

Proof. Let $\varepsilon > 0$, $\delta = \varepsilon$, $L = 4$. If $0 < |x - 2| < \delta$ then $\left| \dfrac{x^2 - 4}{x - 2} - 4 \right| = |x - 2| < \varepsilon$. Hence $f'(2) = 4$.

ANSWER $f'(2) = 4$ **(δ = ε; e.g. ε = 0.01 needs δ = 0.01).**

Marks: definition 1 · scratch 1 · δ and conclusion 1.

2 CALC-24 Q5 · related FYUGP

$f'(c)$ for $f(x) = 3x$ from the definition.

KNOW FIRST

T01 The derivative: a limit of chord slopes M01 Limit of a difference quotient

For $x \neq c$: $\dfrac{f(x) - f(c)}{x - c} = \dfrac{3x - 3c}{x - c} = 3$. This is the constant 3, so for every ε > 0 any δ (say δ = 1) gives $|3 - 3| = 0 < \varepsilon$.

ANSWER $f'(c) = 3$ **for every** $c \in \mathbb{R}$.

Marks: definition 1 · quotient 1 · conclusion 1.

3

Derivative of $g(x) = 1/x$ at $c \neq 0$ from the definition.

KNOW FIRST

T01 The derivative: a limit of chord slopes M02 Factorising and rationalising M04 Limit laws and the squeeze

For $x \neq 0$, $x \neq c$: $\dfrac{1/x - 1/c}{x - c} = \dfrac{c - x}{xc\,(x - c)} = \dfrac{-1}{xc}$. As $x \to c$ the limit laws (denominator $\to c^2 \neq 0$) give $-\dfrac{1}{c^2}$.

ANSWER $g'(c) = -\dfrac{1}{c^2}$, $c \neq 0$. **Check c = 2: quotient at x = 2.001 is** $-1/(2.001 \cdot 2) = -0.24988 \approx -0.25$.

Marks: quotient 1 · cancel $x - c$ 1 · limit 1.

4

$f'(c) = \lim n\{f(c+1/n) - f(c)\}$; the converse fails for $|x|$.

KNOW FIRST

T01 The derivative: a limit of chord slopes P05 Sequences and the useful ones M03 One-sided limits and one-sided derivatives

Let $x_n = c + \dfrac{1}{n}$. Then $x_n \to c$ and $x_n \neq c$, and $\dfrac{f(x_n) - f(c)}{x_n - c} = n\{f(c + \dfrac{1}{n}) - f(c)\}$. By the sequential criterion for limits, since the quotient tends to $f'(c)$ as $x \to c$, this sequence converges to $f'(c)$.

Converse false. $f(x) = |x|$, c = 0: $n\{|\frac{1}{n}| - 0\} = 1$ for all n, so the sequence converges to 1. But $f'(0)$ does not exist: the quotient is $+1$ for $x > 0$ and -1 for $x < 0$ (e.g. $x_n = -1/n$ gives -1).

Marks: sequence and criterion 2 · counterexample 1.

5

$f(x) = x^{1/3}$ is continuous but not differentiable at 0.

KNOW FIRST

T02 Differentiable ⇒ continuous (and not conversely) P05 Sequences and the useful ones

The cube root is continuous at 0 (composition of continuous functions; or given ε, δ = ε³ works). For $x \neq 0$: $\dfrac{f(x) - f(0)}{x - 0} = x^{-2/3}$. Take $x_n = n^{-3} \to 0$:

the quotient is $n^2 \to \infty$ (n = 10: 100). The quotient is unbounded, so it has no finite limit.

ANSWER $f'(0)$ **does not exist (vertical tangent); f is continuous at 0.**

Marks: continuity 1 · quotient 1 · sequence and conclusion 1.

6

State the quotient rule; differentiate $x/(1+x^2)$**.**

KNOW FIRST

T03 Algebra of derivatives: sum, product, quotient

P02 Plus Two rules you may quote

Theorem 6.1.3(d). If f and g are differentiable at c and $g(c) \neq 0$, then f/g is differentiable at c and $\left(\frac{f}{g}\right)'(c) = \frac{f'(c)g(c) - f(c)g'(c)}{(g(c))^2}$.

Here $f(x) = x$, $g(x) = 1 + x^2 \geq 1 \neq 0$:
$\frac{1 \cdot (1+x^2) - x \cdot 2x}{(1+x^2)^2}$.

ANSWER $\left(\frac{x}{1+x^2}\right)' = \frac{1-x^2}{(1+x^2)^2}$**. Check x = 0: 1; x = 1: 0.**

Marks: statement 1 · f, g, g ≠ 0 1 · result 1.

7

Induction: $D\,x^n = nx^{n-1}$**; deduce** $D\,x^{-n}$**.**

KNOW FIRST

T04 Sums, products of many functions; the power rule **T03** Algebra of derivatives: sum, product, quotient

Base. n = 1: $\dfrac{x-c}{x-c} = 1 = 1 \cdot x^0$.

Step. Assume $Dx^k = kx^{k-1}$. Product rule on $x^{k+1} = x^k \cdot x$: $kx^{k-1} \cdot x + x^k \cdot 1 = (k+1)x^k$.

Negative powers. For $x \neq 0$, $x^{-n} = 1/x^n$, $x^n \neq 0$. Quotient rule: $\dfrac{0 \cdot x^n - 1 \cdot nx^{n-1}}{x^{2n}} = -nx^{-n-1}$.

ANSWER $Dx^n = nx^{n-1}$ **(**$n \in \mathbb{N}$**);** $Dx^{-n} = -nx^{-n-1}$ **(**$x \neq 0$**).**

Marks: base 1 · inductive step 1 · negative powers 1.

8

State the chain rule; differentiate $\sqrt{5 - 2x + x^2}$; why differentiable on $\mathbb{R}$.

KNOW FIRST

T06 The chain rule M08 Composite bookkeeping

Theorem 6.1.6 (Chain Rule). Let I, J be intervals in $\mathbb{R}$, let $g : I \to \mathbb{R}$ and $f : J \to \mathbb{R}$ be functions such that $f(J) \subseteq I$, and let $c \in J$. If f is differentiable at c and g is differentiable at $f(c)$, then $g \circ f$ is differentiable at c and $(g \circ f)'(c) = g'(f(c)) \cdot f'(c)$.

Inner $f(x) = x^2 - 2x + 5 = (x-1)^2 + 4 \geq 4 > 0$; outer $g(y) = \sqrt{y}$, differentiable for every $y > 0$, $g'(y) = \dfrac{1}{2\sqrt{y}}$. Since f takes only values in $(0, \infty)$, the rule applies at every x.

ANSWER $g'(x) = \dfrac{2x-2}{2\sqrt{x^2-2x+5}} = \dfrac{x-1}{\sqrt{x^2-2x+5}}$.
Check x = 3: $2/\sqrt{8} = 0.7071$.

Marks: statement 1 · f, g and positivity 1 · derivative 1.

9

Even f has odd f'; odd f has even f'.

KNOW FIRST

T06 The chain rule M08 Composite bookkeeping

Let $h(x) = f(-x)$. Chain rule with inner $x \mapsto -x$ (derivative −1): $h'(x) = -f'(-x)$.

f even: $h = f$, so $f'(x) = -f'(-x)$, i.e. $f'(-x) = -f'(x)$: f' is odd.

f odd: $h = -f$, so $-f'(x) = -f'(-x)$, i.e. $f'(-x) = f'(x)$: f' is even. ∎

Check: x^2 is even, $2x$ is odd; $\sin$ is odd, $\cos$ is even.

Marks: chain rule 1 · even case 1 · odd case 1.

10

$|f|'(c)$ when $f(c) \neq 0$ and when $f(c) = 0$; $|x^2 - 1|$.

KNOW FIRST

T07 The derivative of |f| and of sgn T06 The chain rule M03 One-sided limits and one-sided derivatives

f(c) ≠ 0. $|y|$ is differentiable at $y \neq 0$ with derivative $\operatorname{sgn} y$; by the chain rule $|f|'(c) = \operatorname{sgn}(f(c))\, f'(c)$.

f(c) = 0. $\dfrac{|f(x)|}{x - c} = \pm \left|\dfrac{f(x)}{x - c}\right|$ with the sign of $x - c$, and $\left|\dfrac{f(x)}{x - c}\right| \to |f'(c)|$. Right limit $|f'(c)|$, left limit

$-|f'(c)|$: equal iff $f'(c) = 0$. So $|f|$ is differentiable at c iff $f'(c) = 0$.

ANSWER $f(x) = x^2 - 1$: $|f|'(x) = 2x \operatorname{sgn}(x^2 - 1) = 2x$ **for** $|x| > 1$, $-2x$ **for** $|x| < 1$; **fails at** $x = \pm 1$ **since** $f'(\pm 1) = \pm 2 \neq 0$.

Check: x = 2: 4; x = 0.5: $|f| = 1 - x^2$, derivative −1 ✓.

Marks: f(c) ≠ 0 1 · f(c) = 0 1 · example 1.

11

$h(x) = x|x|$ is differentiable on $\mathbb{R}$.

KNOW FIRST

T07 The derivative of |f| and of sgn M03 One-sided limits and one-sided derivatives

$x > 0$: $h = x^2$, $h' = 2x$. $x < 0$: $h = -x^2$, $h' = -2x$. At 0: $\dfrac{h(x) - h(0)}{x - 0} = \dfrac{x|x|}{x} = |x| \to 0$ from both sides.

ANSWER $h'(x) = 2|x|$ **for all** $x \in \mathbb{R}$ **(continuous; but** $h''(0)$ **does not exist).**

Marks: x ≠ 0 1 · x = 0 quotient 1 · answer 1.

12

Prove $D \tan x = \sec^2 x$ **and** $D \sec x = \sec x \tan x$**; domain.**

KNOW FIRST

T08 Derivatives of tan, sec, cot, csc T03 Algebra of derivatives: sum, product, quotient P02 Plus Two rules you may quote

On $\cos x \neq 0$, i.e. $x \neq (2k+1)\pi/2$, use the quotient rule with $\sin' = \cos$, $\cos' = -\sin$:

$$(\tan x)' = \frac{\cos x \cos x - \sin x(-\sin x)}{\cos^2 x} = \frac{1}{\cos^2 x} = \sec^2 x.$$

$$(\sec x)' = \left(\frac{1}{\cos x}\right)' = \frac{0 \cdot \cos x - 1 \cdot (-\sin x)}{\cos^2 x} = \frac{\sin x}{\cos x} \cdot \frac{1}{\cos x} = \sec x \tan x.$$

ANSWER **Both valid for $x \neq (2k+1)\pi/2$, $k \in \mathbb{Z}$. Check x = π/4: $\sec^2 = 2$ ✓.**

Marks: quotient setup 1 · tan 1 · sec and domain 1.

13

For which $r > 0$ does $f'(0)$ exist, $f(x) = x^r \sin(1/x)$, $f(0) = 0$?

KNOW FIRST

T09 x² sin(1/x): differentiable, but f′ is not continuous M04 Limit laws and the squeeze M07 No limit, by two sequences

Quotient at 0: $\dfrac{f(x) - f(0)}{x} = |x|^{r-1} \sin(1/x)$ (take $x > 0$, or $|x|^r$ for $x < 0$).

r > 1. $|x|^{r-1} \sin(1/x)| \leq |x|^{r-1}$. Given ε > 0 take $\delta = \varepsilon^{1/(r-1)}$: then $0 < |x| < \delta$ gives quotient $< \varepsilon$. So $f'(0) = 0$.

r = 1. Quotient is $\sin(1/x)$. $x_n = \dfrac{1}{2n\pi} \to 0$ gives 0; $y_n = \dfrac{1}{2n\pi + \pi/2} \to 0$ gives $\sin(2n\pi + \pi/2) = 1$. No limit.

0 < r < 1. At $y_n = 1/(2n\pi + \pi/2)$: quotient $= y_n^{r-1} \to \infty$ (r = ½, n = 100: ≈ 25.1). Unbounded, no limit.

ANSWER $f'(0)$ **exists iff** $r > 1$**, and then** $f'(0) = 0$ **.**

Marks: quotient 1 · r > 1 with δ 1 · r ≤ 1 1.

14 CALC-24 Q10 · related FYUGP

Critical points of $x^3 - 3x + 1$; which can be extrema.

KNOW FIRST

T10 Relative extrema; the Interior Extremum Theorem P02 Plus Two rules you may quote

Theorem 6.2.1 (Interior Extremum). Let c be an interior point of the interval I at which $f : I \to \mathbb{R}$ has a relative extremum. If the derivative of f at c exists, then $f'(c) = 0$.

$f'(x) = 3x^2 - 3 = 3(x-1)(x+1)$; f' exists everywhere on $\mathbb{R}$ (every point is interior). So any relative extremum has $f'(c) = 0$: $c = \pm 1$. These are only *candidates* (converse false: x^3 at 0).

Sign of f': $+$ on $(-\infty, -1)$, $-$ on $(-1, 1)$, $+$ on $(1, \infty)$, so −1 is a relative maximum and 1 a relative minimum.

ANSWER Critical points $x = -1, 1$**;** $f(-1) = 3$ **(max),** $f(1) = -1$ **(min).**

Marks: theorem 1 · f′ = 0 1 · classification 1.

15 CALC-24 Q18 · related FYUGP

Relative extrema of $f(x) = \dfrac{x^2}{x^4 + 16}$ by the First Derivative Test.

KNOW FIRST

T15 The First Derivative Test T10 Relative extrema; the Interior Extremum Theorem M05 Sign analysis of the derivative

$$f'(x) = \frac{2x(x^4 + 16) - x^2 \cdot 4x^3}{(x^4 + 16)^2} = \frac{2x(16 - x^4)}{(x^4 + 16)^2}.$$

Zeros: $x = 0, \pm 2$ (denominator > 0).

INTERVAL	$X < -2$	$-2 < X < 0$	$0 < X < 2$	$X > 2$
f'	+	–	+	–

By the First Derivative Test (6.2.8): $f' \geq 0 \to f' \leq 0$ at ±2 gives maxima, $f' \leq 0 \to f' \geq 0$ at 0 a minimum.

ANSWER **Maxima $f(\pm 2) = \dfrac{4}{32} = \dfrac{1}{8}$; minimum $f(0) = 0$.**

Check: f(1) = 1/17 = 0.059 < 0.125; f(3) = 9/97 = 0.093 < 0.125.

Marks: derivative 1 · sign chart 1 · conclusion with values 1.

16

Verify the MVT for $f(x) = x^3$ on $[0, 2]$; find c.

KNOW FIRST

T12 The Mean Value Theorem

Hypotheses: f is a polynomial, so continuous on $[0, 2]$ and differentiable on $(0, 2)$. Chord slope

$\frac{f(2)-f(0)}{2-0} = \frac{8}{2} = 4$. Solve $f'(c) = 3c^2 = 4$: $c = \pm\frac{2}{\sqrt{3}}$; only the positive root lies in $(0, 2)$.

ANSWER $c = \frac{2}{\sqrt{3}} \approx 1.1547 \in (0, 2)$. **Check:** $3c^2 = 3 \cdot \frac{4}{3} = 4$ ✓.

Marks: hypotheses 1 · equation 1 · c in the interval 1.

17 CALC-24 Q9 · related FYUGP

Define increasing and strictly increasing; example with a vanishing derivative.

KNOW FIRST

T14 Increasing and decreasing: the sign of f′ P03 Graph shapes to recognise

Definition. $f : I \to \mathbb{R}$ is **increasing** on I if $x_1 < x_2$ in I implies $f(x_1) \le f(x_2)$; **strictly increasing** if $x_1 < x_2$ implies $f(x_1) < f(x_2)$. (In Bartle's sense a constant function is increasing.)

Example. $f(x) = x^3$: $x_1 < x_2 \Rightarrow x_1^3 < x_2^3$, so strictly increasing; yet $f'(0) = 0$. (So "$f' > 0$" is sufficient for strict increase, not necessary.)

Marks: both definitions 1.5 · example 1.5.

18 CALC-24 Q20(1) · related FYUGP

Intervals of increase and decrease of $x^4 - 5x^3 + 9x^2$.

KNOW FIRST

T14 Increasing and decreasing: the sign of f′ M05 Sign analysis of the derivative P13 Quadratics and sign charts

$f'(x) = 4x^3 - 15x^2 + 18x = x(4x^2 - 15x + 18)$. The quadratic has discriminant $225 - 4 \cdot 4 \cdot 18 = 225 - 288 = -63 < 0$ and leading coefficient 4 > 0, so it is positive for every x. Hence $\operatorname{sign} f' = \operatorname{sign} x$.

By Theorem 6.2.7: $f' \le 0$ on $(-\infty, 0]$ gives decreasing; $f' \ge 0$ on $[0, \infty)$ gives increasing.

ANSWER **Decreasing on $(-\infty, 0]$, increasing on $[0, \infty)$; minimum $f(0) = 0$. Check: f(−1) = 15, f(0) = 0, f(1) = 5.**

Marks: derivative 1 · quadratic sign 1 · intervals 1.

19 MTS-22 MCQ 5 · adapted

Which of $1 - x^3$, $x^2 + x$, $1/x^2$ increase on $(0, 1)$?

KNOW FIRST

T14 Increasing and decreasing: the sign of f′ M05 Sign analysis of the derivative

On $(0, 1)$: $(1 - x^3)' = -3x^2 < 0$; $(x^2 + x)' = 2x + 1 > 0$; $(1/x^2)' = -2/x^3 < 0$. By 6.2.7 (and its strict form), $f' > 0$ gives strictly increasing, $f' < 0$ decreasing.

ANSWER **Only (ii) $x^2 + x$ is increasing on $(0, 1)$. Check: values at 0.1 and 0.5: 0.11 → 0.75 ✓; 0.999 → 0.875 and 100 → 4 decrease.**

Marks: derivatives 1.5 · conclusions 1.5.

20

State Lemma 6.2.11 and prove part (a).

KNOW FIRST

T22 The local sign lemma P04 Limits of functions (Real Analysis I)

Lemma 6.2.11. Let $f : I \to \mathbb{R}$ have a derivative at $c \in I$. (a) If $f'(c) > 0$, then there is $\delta > 0$ such that $f(x) > f(c)$ for $x \in I$ with $c < x < c + \delta$. (b) If $f'(c) < 0$, then there is $\delta > 0$ such that $f(x) > f(c)$ for $x \in I$ with $c - \delta < x < c$.

Proof of (a). Put $\varepsilon = f'(c)/2 > 0$. By the definition of $f'(c)$ there is $\delta > 0$ with $\left| \dfrac{f(x) - f(c)}{x - c} - f'(c) \right| < \dfrac{f'(c)}{2}$ for $x \in I$, $0 < |x - c| < \delta$; hence the quotient is $> \dfrac{f'(c)}{2} > 0$. For $c < x < c + \delta$, $x - c > 0$, so $f(x) - f(c) = (x - c) \cdot (\text{quotient}) > 0$. ∎

For (b): same δ, but now $x - c < 0$ and quotient < 0, product > 0.

Marks: statement 1 · δ from ε = f′(c)/2 1 · sign of product 1.

Section B · Paragraph / problem
6 marks each.

21 Bartle Thm 6.1.2 related MAT-22 MCQ 4, MAT-21 MCQ 3

Differentiable at c implies continuous at c; the converse is false.

KNOW FIRST

T02 Differentiable ⇒ continuous (and not conversely) T01 The derivative: a limit of chord slopes P06 Continuity (Module 1) M04 Limit laws and the squeeze

Theorem 6.1.2. If $f : I \to \mathbb{R}$ has a derivative at $c \in I$, then f is continuous at c.

SKELETON

Want $f(x) \to f(c)$, i.e. $f(x) - f(c) \to 0$. Write $f(x) - f(c) = \dfrac{f(x) - f(c)}{x - c} \cdot (x - c)$ and use the limit of a product: $f'(c) \cdot 0$?

Proof. For $x \in I$, $x \neq c$:

$$f(x) - f(c) = \frac{f(x) - f(c)}{x - c} \cdot (x - c).$$

As $x \to c$ the first factor tends to $f'(c)$ (hypothesis) and the second to 0. By the limit theorem for products, $\lim_{x \to c} \big(f(x) - f(c)\big) = f'(c) \cdot 0 = 0$, so $\lim_{x \to c} f(x) = f(c)$. This is continuity at c. ∎

Converse false. $f(x) = |x|$ is continuous at 0 ($||x| - 0| = |x - 0|$, so δ = ε). But $\dfrac{|x| - 0}{x - 0} = +1$ for

$x > 0$ and -1 for $x < 0$: with $x_n = 1/n$ the quotients are 1, with $y_n = -1/n$ they are -1, both $\to 0$. The limits differ, so $f'(0)$ does not exist.

ANSWER **Differentiable ⇒ continuous; $|x|$ is continuous at 0 but not differentiable there.**

Marks: statement 1 · factor identity 1 · limit product 2 · conclusion 1 · example 1.

22 CALC-24 Q13 · related FYUGP

$f = x^2$ ($x \le 1$), $4\sqrt{x}$ ($x > 1$): differentiability at 1; the fixed-up F and G.

KNOW FIRST

T02 Differentiable ⇒ continuous (and not conversely) T01 The derivative: a limit of chord slopes M03 One-sided limits and one-sided derivatives P10 Piecewise functions and the signum

(a) $f(1) = 1$, but $\lim_{x \to 1^+} 4\sqrt{x} = 4 \neq 1$. So f is **not continuous** at 1 (e.g. $x_n = 1 + \dfrac{1}{n}$: $f(x_n) = 4\sqrt{1 + 1/n} \to 4$). By Theorem 6.1.2 (contrapositive) f is not differentiable at 1. Directly: $\dfrac{4\sqrt{x} - 1}{x - 1}$ at $x = 1.01$ is $\dfrac{4.0200 - 1}{0.01} \approx 302$ and grows without bound. (The left derivative alone would be 2: the "find the value" is a trap.)

(b) F must be continuous at 1: $a \cdot 1 + b = 1$. Left derivative $= 2 \cdot 1 = 2$; right derivative $= a$. So $a = 2$, $b = -1$, $F(x) = 2x - 1$ for $x > 1$ (the tangent line of x^2 at 1), and $F'(1) = 2$.

(c) $G(1) = 1 = \lim_{x\to 1^+}\sqrt{x}$, so G is continuous at 1. Left derivative $\lim \dfrac{x^2-1}{x-1} = 2$. Right derivative $\lim \dfrac{\sqrt{x}-1}{x-1} = \lim \dfrac{1}{\sqrt{x}+1} = \dfrac{1}{2}$. Since $2 \neq \dfrac{1}{2}$, G is not differentiable at 1.

ANSWER **(a) No. (b) $a = 2$, $b = -1$. (c) No (left 2, right ½).**

Marks: (a) discontinuity and theorem 2 · (b) 2 · (c) 2.

23 Bartle Thm 6.1.3(c)

State and prove the product rule; deduce $(1/g)'$.

KNOW FIRST

T03 Algebra of derivatives: sum, product, quotient **T02** Differentiable ⇒ continuous (and not conversely) **T01** The derivative: a limit of chord slopes

Theorem 6.1.3(c). Let $f, g : I \to \mathbb{R}$ be differentiable at $c \in I$. Then the product fg is differentiable at c and $(fg)'(c) = f'(c)g(c) + f(c)g'(c)$.

Proof. For $x \neq c$, adding and subtracting $f(c)g(x)$:

$$\frac{f(x)g(x) - f(c)g(c)}{x-c} = \frac{f(x)-f(c)}{x-c}\,g(x) + f(c)\,\frac{g(x)}{x}$$

Since g is differentiable at c it is continuous at c (Thm 6.1.2), so $g(x) \to g(c)$. By the limit laws the right side tends to $f'(c)g(c) + f(c)g'(c)$. ∎

Reciprocal. Let $g(c) \neq 0$. g is continuous at c, so $g \neq 0$ on a neighbourhood of c (local sign, Thm 4.2.9) and $1/g$ is defined there. For x in it, $x \neq c$:

$$\frac{1/g(x)-1/g(c)}{x-c}=\frac{-1}{g(x)g(c)}\cdot\frac{g(x)-g(c)}{x-c}\longrightarrow\frac{-g'(c)}{g(c)^2}$$

ANSWER $(1/g)'(c)=-\dfrac{g'(c)}{g(c)^2}$. **Check $g=x$, c = 2:** $-1/4$, **and** $(1/x)'=-1/x^2=-0.25$ ✓.

Marks: statement 1 · add-and-subtract identity 2 · continuity of g 1 · reciprocal 2.

24 Bartle Thm 6.1.5

Carathéodory's Theorem; φ for x^3; none for $|x|$.

KNOW FIRST

T05 Carathéodory's characterisation T01 The derivative: a limit of chord slopes P06 Continuity (Module 1) M09 Defining a function by cases, and the join

Theorem 6.1.5 (Carathéodory). Let f be defined on an interval I containing the point c. Then f is differentiable at c if and only if there exists a function φ on I that is continuous at c and satisfies $f(x)-f(c)=\varphi(x)(x-c)$ for $x\in I$. In this case $\varphi(c)=f'(c)$.

1. **(⇒)** Define $\varphi(x)=\dfrac{f(x)-f(c)}{x-c}$ for $x\neq c$, and $\varphi(c)=f'(c)$.
   The only possible candidate: the difference quotient, completed at c.

2. $\lim_{x\to c}\varphi(x)=f'(c)=\varphi(c)$, so φ is continuous at c.
   Definition of $f'(c)$.

3. For $x \neq c$ multiply by $x - c$; for $x = c$ both sides are 0. So $f(x) - f(c) = \varphi(x)(x - c)$ on all of I.
   Check every x, including c.

4. **(⇐)** Given such a φ: for $x \neq c$, $\dfrac{f(x) - f(c)}{x - c} = \varphi(x)$.
   Divide by $x - c \neq 0$.

5. φ is continuous at c, so $\lim_{x \to c} \varphi(x) = \varphi(c)$. Hence $f'(c)$ exists and equals $\varphi(c)$. ∎
   Definition 6.1.1.

$f(x) = x^3$. $x^3 - c^3 = (x^2 + cx + c^2)(x - c)$, so $\varphi(x) = x^2 + cx + c^2$, which is continuous everywhere, and $f'(c) = \varphi(c) = 3c^2$.

$|x|$ **at 0.** $|x| - 0 = \varphi(x) \cdot x$ forces $\varphi(x) = |x|/x = \operatorname{sgn} x$ for $x \neq 0$; its limits at 0 are $+1$ (right) and -1 (left), so no value $\varphi(0)$ makes φ continuous. Hence $|x|$ is not differentiable at 0.

Marks: statement 1 · (⇒) 2 · (⇐) 1 · x^3 1 · $|x|$ 1.

25 Bartle Ex 6.1.7(e)

$x^2 \sin(1/x)$: differentiable everywhere, f' not continuous at 0.

KNOW FIRST

T09 $x^2 \sin(1/x)$: differentiable, but f′ is not continuous
T01 The derivative: a limit of chord slopes M04 Limit laws and the squeeze M07 No limit, by two sequences

$x \neq 0$. Product and chain rules: $f'(x) = 2x\sin\dfrac{1}{x} + x^2\cos\dfrac{1}{x}\cdot\left(-\dfrac{1}{x^2}\right) = 2x\sin\dfrac{1}{x} - \cos\dfrac{1}{x}$.

$x = 0$ (the rules cannot be used: $\sin(1/x)$ is undefined at 0). From the definition: $\dfrac{f(x)-f(0)}{x-0} = x\sin\dfrac{1}{x}$, and $|x\sin\dfrac{1}{x}| \le |x| \to 0$ (δ = ε). So $f'(0) = 0$.

f' not continuous at 0. $x_n = \dfrac{1}{2n\pi} \to 0$: $f'(x_n) = 0 - \cos(2n\pi) = -1$. $y_n = \dfrac{1}{(2n+1)\pi} \to 0$: $f'(y_n) = 0 - \cos((2n+1)\pi) = +1$. The limits $-1 \neq +1$, and neither equals $f'(0) = 0$, so f' has no limit at 0.

ANSWER $f'(x) = 2x\sin\dfrac{1}{x} - \cos\dfrac{1}{x}$ **($x \neq 0$), $f'(0) = 0$; f' is discontinuous at 0.**

Numerical check: n = 1: $f'(1/2\pi) = -1$, $f'(1/3\pi) = +1$.

Marks: x ≠ 0 1 · $f'(0)$ with squeeze 2 · two sequences 2 · conclusion 1.

26 Bartle Thm 6.2.1

Interior Extremum Theorem; "interior", "differentiable", converse.

KNOW FIRST

T10 Relative extrema; the Interior Extremum Theorem T22 The local sign lemma P08 Intervals, neighbourhoods and interior points

Definition 6.2.1. f has a **relative maximum** at $c \in I$ if there is a neighbourhood $V = V_\delta(c)$ with $f(x) \le$

$f(c)$ for all $x \in V \cap I$ (relative minimum: $f(c) \leq f(x)$).

Theorem 6.2.1. Let c be an interior point of the interval I at which $f : I \to \mathbb{R}$ has a relative extremum. If the derivative of f at c exists, then $f'(c) = 0$.

> **SKELETON**
>
> Max case; min by applying it to $-f$. Assume $f'(c) \neq 0$ and produce points arbitrarily near c with $f(x) > f(c)$? (contradiction).

Proof (relative maximum at c). Suppose $f'(c) > 0$. Put $\varepsilon = f'(c)/2$; there is $\delta_1 > 0$ with $\dfrac{f(x) - f(c)}{x - c} > \dfrac{f'(c)}{2} > 0$ for $0 < |x - c| < \delta_1$. Because c is *interior* we may choose $\delta \leq \min\{\delta_1, \delta_0\}$ (δ_0 the radius of the maximum neighbourhood) with $(c - \delta, c + \delta) \subseteq I$. Take x with $c < x < c + \delta$: then $f(x) - f(c) = (x - c)(\text{positive}) > 0$, contradicting $f(x) \leq f(c)$.

Suppose $f'(c) < 0$: the quotient is < 0 near c, so for $c - \delta < x < c$ we have $(x - c) < 0$ and $f(x) - f(c) > 0$, again a contradiction. Hence $f'(c) = 0$. ∎

Hypotheses. Interior: $f(x) = x$ on $[0, 1]$ has extrema at 0 and 1, but $f' = 1$. Differentiable: $|x|$ on $[-1, 1]$ has a minimum at 0 where $f'(0)$ does not exist. Converse: $f(x) = x^3$ has $f'(0) = 0$ but no extremum at 0 (negative to the left, positive to the right).

Marks: definition and statement 1 · contradiction setup 1 · f′ > 0 2 · f′ < 0 1 · examples 1.

27 Bartle Thm 6.2.3

Rolle's Theorem; verify for $x(x-1)(x-2)$ on $[0,2]$.

KNOW FIRST

T11 Rolle's Theorem T10 Relative extrema; the Interior Extremum Theorem P07 The big theorems of Module 1

Theorem 6.2.3 (Rolle). Suppose that f is continuous on a closed interval $I=[a,b]$, that the derivative f' exists at every point of the interior (a,b), and that $f(a)=f(b)=0$. Then there exists at least one point c in (a,b) such that $f'(c)=0$.

1. If $f\equiv 0$ on $[a,b]$, any $c\in(a,b)$ works.
   $f'=0$ everywhere.

2. Otherwise f takes a nonzero value; replacing f by $-f$ (which has the same critical points) we may assume f takes a positive value.
   $(-f)'(c)=0 \iff f'(c)=0.$

3. f is continuous on the closed bounded interval, so it attains a maximum $M>0$ at some $c\in[a,b]$.
   Max–Min Theorem 5.3.4.

4. $f(a)=f(b)=0<M$, so $c\neq a,b$: $c\in(a,b)$, and $f'(c)$ exists.
   Differentiable on the open interval.

5 f has a relative maximum at the interior point c, hence $f'(c) = 0$. ∎

Interior Extremum Theorem 6.2.1.

Verification. $f(x) = x^3 - 3x^2 + 2x$ is a polynomial; $f(0) = f(1) = f(2) = 0$. $f'(x) = 3x^2 - 6x + 2 = 0$ gives $x = 1 \pm \dfrac{1}{\sqrt{3}}$.

ANSWER $c_1 = 1 - \dfrac{1}{\sqrt{3}} \approx 0.4226$, $c_2 = 1 + \dfrac{1}{\sqrt{3}} \approx 1.5774$, **both in** $(0, 2)$; $f(c_{1,2}) = \pm\dfrac{2}{3\sqrt{3}} \approx \pm 0.3849$ **. c need not be unique.**

Marks: statement 1 · reduction 1 · max attained and interior 2 · conclusion 1 · verification 1.

28 Bartle Thm 6.2.5, Cor 6.2.6

$f' = 0$ implies constant; equal derivatives differ by a constant.

KNOW FIRST

T13 Zero derivative means constant T12 The Mean Value Theorem

Theorem 6.2.5. Suppose f is continuous on the closed interval $I = [a, b]$, differentiable on (a, b), and $f'(x) = 0$ for $x \in (a, b)$. Then f is constant on I.

Proof. Fix $x \in (a, b]$. f is continuous on $[a, x]$ and differentiable on (a, x), so by the Mean Value Theorem there is $c \in (a, x)$ with $f(x) - f(a) = f'(c)(x - a) = 0 \cdot (x - a) = 0$. So $f(x) = f(a)$ for every $x \in I$. ∎

Corollary 6.2.6. If f, g are continuous on $[a,b]$, differentiable on (a,b) and $f' = g'$ there, then $f = g + C$. *Proof.* $h = f - g$ is continuous, differentiable, $h' = f' - g' = 0$ on (a,b); by 6.2.5 $h \equiv C$. ∎

Why an interval is needed. On $(0,1) \cup (2,3)$ let $f = 0$ on $(0,1)$ and $f = 1$ on $(2,3)$. Then $f' = 0$ everywhere but f is not constant. The MVT needs the whole segment $[a,x]$ inside the domain; here the segment from $\frac{1}{2}$ to $\frac{5}{2}$ leaves it.

Example: $\sin^2 x + \cos^2 x$ has derivative $2\sin x \cos x - 2\cos x \sin x = 0$ and equals 1 at x = 0, so it equals 1 everywhere.

Marks: statement 1 · MVT application 2 · conclusion 1 · corollary 1 · counterexample 1.

29 Bartle Thm 6.2.8

First Derivative Test, part (a); $x^3 - 3x$; $|x|$ at 0.

KNOW FIRST

T15 The First Derivative Test T12 The Mean Value Theorem M05 Sign analysis of the derivative

Theorem 6.2.8 (a). Let f be continuous on the interval $I = [a,b]$ and let c be an interior point of I. Assume f is differentiable on (a,c) and (c,b). If there is a neighbourhood $(c-\delta, c+\delta) \subseteq I$ such that $f'(x) \geq 0$ for $c - \delta < x < c$ and $f'(x) \leq 0$ for $c < x < c + \delta$, then f has a relative maximum at c.

Proof. *Left side.* For $c - \delta < x < c$, f is continuous on $[x,c]$ and differentiable on (x,c). MVT: $f(c) - f(x) = f'(\xi)(c - x)$ with $\xi \in (x,c)$, $f'(\xi) \geq 0$, $c -$

$x > 0$; so $f(x) \le f(c)$. *Right side.* For $c < x < c + \delta$, MVT on $[c, x]$: $f(x) - f(c) = f'(\xi)(x - c)$ with $f'(\xi) \le 0$, $x - c > 0$; so $f(x) \le f(c)$. Thus $f(x) \le f(c)$ on $(c - \delta, c + \delta)$: a relative maximum. ∎ ((b) is the same with signs reversed.)

$f = x^3 - 3x$. $f' = 3(x^2 - 1)$: ≥ 0 on $(-\infty, -1)$, ≤ 0 on $(-1, 1)$, ≥ 0 on $(1, \infty)$. Relative maximum at -1 ($f = 2$); relative minimum at 1 ($f = -2$).

$|x|$ at 0. $f' = -1$ for $x < 0$, $+1$ for $x > 0$: part (b) gives a relative minimum at 0, although $f'(0)$ does not exist (so the test does not need $f'(c)$).

Marks: statement 1 · left side 1.5 · right side 1.5 · examples 2.

30 Bartle 6.2.9(a) related MAT-21 Q17

Zero of $\cos$ between zeros of $\sin$; $x^3 + 3x + 1$ has exactly one real root.

KNOW FIRST

T16 Rolle for roots: location, interlacing, counting
T11 Rolle's Theorem P07 The big theorems of Module 1

(a) Zeros of $\sin$ are $k\pi$. On $[k\pi, (k+1)\pi]$, $\sin$ is continuous, differentiable, and $\sin(k\pi) = \sin((k+1)\pi) = 0$. By Rolle's Theorem there is c in $(k\pi, (k+1)\pi)$ with $\sin'(c) = \cos c = 0$ (indeed $c = k\pi + \frac{\pi}{2}$).

(b) At least one root. $p(x) = x^3 + 3x + 1$ is continuous, $p(-1) = -3 < 0 < 1 = p(0)$, so by the Location of Roots Theorem p has a root in $(-1, 0)$.

At most one root. Suppose $p(r_1) = p(r_2) = 0$, $r_1 < r_2$. By Rolle (for $[r_1, r_2]$) some c has $p'(c) = 3c^2 + 3 = 0$, impossible since $p'(x) \geq 3 > 0$. So exactly one root.

Location. $p(-\frac{1}{2}) = -\frac{1}{8} - \frac{3}{2} + 1 = -\frac{5}{8} < 0$ and $p(-\frac{1}{4}) = -\frac{1}{64} - \frac{3}{4} + 1 = \frac{15}{64} > 0$.

ANSWER **Exactly one real root, in $(-\frac{1}{2}, -\frac{1}{4})$ (length $\frac{1}{4}$); numerically ≈ -0.3222.**

Marks: (a) 2 · existence 1 · uniqueness by Rolle 2 · location 1.

31 Bartle 6.2.9(b)

MVT for $\sqrt{x}$ on $[100, 105]$: bounds for $\sqrt{105}$.

KNOW FIRST

T17 MVT for approximate calculation T12 The Mean Value Theorem M06 An inequality from the MVT

$f(x) = \sqrt{x}$ is continuous on $[100, 105]$, differentiable on $(100, 105)$, $f'(x) = \frac{1}{2\sqrt{x}}$. MVT: there is $c \in (100, 105)$ with

$$\sqrt{105} - 10 = \frac{105 - 100}{2\sqrt{c}} = \frac{5}{2\sqrt{c}}.$$

First bounds. $10 < \sqrt{c} < \sqrt{105} < \sqrt{121} = 11$, so $\frac{5}{22} < \sqrt{105} - 10 < \frac{5}{20}$, i.e. $10.2272 < \sqrt{105} < 10.25$ (5/22 = 0.22727).

Sharpen. $10.25^2 = 105.0625 > 105$, so $\sqrt{105} < 10.25$ and hence $\sqrt{c} < 10.25$. Then $\sqrt{105} - 10 > \frac{5}{2 \cdot 10.25} = \frac{5}{20.5} = 0.243902\ldots$

ANSWER $10.2439 < \sqrt{105} < 10.25$. **True value** $\sqrt{105} = 10.24695$ ✓.

Marks: MVT 1 · setup and hypotheses 1 · first bounds 2 · sharpening 2.

32 Bartle 6.2.10(a), Ex 6.2.7

$e^x \geq 1 + x$; $\dfrac{x-1}{x} < \ln x < x - 1$ for $x > 1$.

KNOW FIRST

T18 Inequalities from the MVT: exponential and logarithm T12 The Mean Value Theorem M06 An inequality from the MVT P12 Exponential and logarithm

(a) $x = 0$: equality. $x > 0$: MVT for e^t on $[0, x]$: $e^x - 1 = e^c\, x$ with $0 < c < x$. Since $e^c > 1$ and $x > 0$, $e^x - 1 > x$.

$x < 0$: MVT on $[x, 0]$: $1 - e^x = e^c(0 - x) = -x\, e^c$ with $x < c < 0$, so $e^c < 1$ and $-x > 0$: $1 - e^x < -x$, i.e. $e^x > 1 + x$. So $e^x \geq 1 + x$ with equality only at 0.

(b) $x > 1$: MVT for $\ln t$ on $[1, x]$: $\ln x - \ln 1 = \dfrac{1}{c}(x - 1)$ with $1 < c < x$. Then $\dfrac{1}{x} < \dfrac{1}{c} < 1$; multiply by $x - 1 > 0$: $\dfrac{x-1}{x} < \ln x < x - 1$.

ANSWER **Checks: x = −1:** $0.368 > 0$; **x = 1:** $2.718 > 2$; **x = 2:** $0.5 < 0.693 < 1$.

Marks: MVT setup 1 · x > 0 1 · x < 0 1.5 · (b) bound on $1/c$ 1.5 · conclusion 1.

33 Bartle 6.2.10(b), Ex 6.2.15 related MTS-22 MCQ 4

$-x \le \sin x \le x$; sin is 1-Lipschitz; bounded derivative implies Lipschitz.

KNOW FIRST

T19 Inequalities from the MVT: sine, and Lipschitz
T12 The Mean Value Theorem M06 An inequality from the MVT

(c) general. Let f be differentiable on the interval I with $|f'(t)| \le K$ for $t \in I$. For $x < u$ in I, f is continuous on $[x, u]$ and differentiable inside, so MVT gives $\xi \in (x, u)$ with

$$|f(x) - f(u)| = |f'(\xi)|\,|x - u| \le K|x - u|.$$

That is the definition of **Lipschitz** (constant K). Such an f is uniformly continuous on I (take δ = ε/K).

(b) $f = \sin$, $f' = \cos$, $|\cos| \le 1$, so $K = 1$: $|\sin x - \sin y| \le |x - y|$ for all x, y.

(a) Take $y = 0$ in (b): $|\sin x| \le |x| = x$ for $x \ge 0$, i.e. $-x \le \sin x \le x$. ∎

ANSWER **Check: x = 1, y = 0.5:** $|\sin 1 - \sin 0.5| = 0.362 \le 0.5$ ✓**; x = π/2:** $1 \le 1.571$ ✓**.**

Marks: MVT with bound 2 · Lipschitz definition 1 · (b) 1 · (a) 1 · remark 1.

34 Bartle 6.2.10(c)

Bernoulli: $(1+x)^\alpha \geq 1+\alpha x$ **for** $\alpha > 1$, $x > -1$.

KNOW FIRST

T20 Bernoulli's Inequality T12 The Mean Value Theorem M06 An inequality from the MVT

Let $h(t) = (1+t)^\alpha$ on $(-1, \infty)$; $h'(t) = \alpha(1+t)^{\alpha-1}$ (standard derivative of a power; rational α in Bartle 6.1.10(d), real α in §8.3). Note $\alpha - 1 > 0$.

$x = 0$: equality $1 = 1$.

$x > 0$: MVT on $[0, x]$: $(1+x)^\alpha - 1 = \alpha(1+c)^{\alpha-1}x$, $0 < c < x$. As $1+c > 1$ and $\alpha - 1 > 0$, $(1+c)^{\alpha-1} > 1$, so $(1+x)^\alpha - 1 > \alpha x$.

$-1 < x < 0$: MVT on $[x, 0]$: $1 - (1+x)^\alpha = \alpha(1+c)^{\alpha-1}(-x)$, $x < c < 0$. Now $0 < 1+c < 1$ so $(1+c)^{\alpha-1} < 1$; with $-x > 0$, $1 - (1+x)^\alpha < -\alpha x$, i.e. $(1+x)^\alpha > 1 + \alpha x$. ∎

ANSWER **Equality iff** $x = 0$. **Check** $\alpha = \frac{3}{2}$: **x = 1:** $2.828 > 2.5$; **x = −0.5:** $0.354 > 0.25$.

Marks: h and h′ 1 · x > 0 2 · x < 0 2 · equality 1.

35 Bartle 6.2.10(d)

$x^\alpha \leq \alpha x + (1-\alpha)$; **weighted AM–GM;** $\sqrt{ab} \leq (a+b)/2$.

KNOW FIRST

T21 Weighted AM–GM inequality T15 The First Derivative Test T20 Bernoulli's Inequality

Let $0 < \alpha < 1$ and $g(x) = \alpha x - x^{\alpha}$ for $x \geq 0$. Then g is continuous on $[0, \infty)$ and $g'(x) = \alpha(1 - x^{\alpha-1})$ for $x > 0$. Since $\alpha - 1 < 0$: for $0 < x < 1$, $x^{\alpha-1} > 1$ so $g' < 0$; for $x > 1$, $x^{\alpha-1} < 1$ so $g' > 0$.

So g decreases on $[0, 1]$ and increases on $[1, \infty)$ (6.2.7); g has its absolute minimum at 1: $g(x) \geq g(1) = \alpha - 1$. That is $\alpha x - x^{\alpha} \geq \alpha - 1$, i.e.

$$x^{\alpha} \leq \alpha x + (1 - \alpha), \qquad x \geq 0,$$

with equality iff $x = 1$.

Deduction. For $a, b > 0$ put $x = a/b$: $(a/b)^{\alpha} \leq \alpha \dfrac{a}{b} + (1 - \alpha)$. Multiply by $b > 0$: $a^{\alpha} b^{1-\alpha} \leq \alpha a + (1 - \alpha) b$, equality iff $a = b$.

ANSWER $\alpha = \dfrac{1}{2}$: $\sqrt{ab} \leq \dfrac{a+b}{2}$. **Check** $\alpha = \dfrac{1}{3}$, **a = 8, b = 1:** $2 \leq \dfrac{8}{3} + \dfrac{2}{3} = \dfrac{10}{3}$ ✓.

Marks: g and g′ 1 · sign analysis 2 · minimum and inequality 1 · deduction 1 · AM–GM 1.

36 Bartle Cor 6.2.13, Ex 6.2.12

A function that is not a derivative; two functions with derivative h.

KNOW FIRST

T24 A function that is not a derivative T23 Darboux's Theorem

Darboux's Theorem 6.2.12. If f is differentiable on $I = [a, b]$ and k is a number between $f'(a)$ and $f'(b)$, then there is at least one point c in (a, b) with $f'(c) = k$.

g is not a derivative. Suppose $g = f'$ on $[-1, 1]$ for some differentiable f. Then $f'(-1) = -1 < \frac{1}{2} < 1 = f'(1)$, so by Darboux there is $c \in (-1, 1)$ with $g(c) = \frac{1}{2}$. But g takes only the values $-1, 0, 1$. Contradiction.

Two functions. $f_1(x) = \max\{x, 0\}$ ($= 0$ for $x \le 0$, x for $x > 0$) and $f_2(x) = 0$ for $x \le 0$, $f_2(x) = x + 1$ for $x > 0$. For $x < 0$ both have derivative 0; for $x > 0$ both have derivative 1; so both have derivative $h(x)$ at every $x \neq 0$. But $f_2 - f_1 = 0$ on $(-\infty, 0]$ and $= 1$ on $(0, \infty)$: not constant.

ANSWER **Equal derivatives off 0 need not give a constant difference, because $\mathbb{R} \setminus \{0\}$ is not an interval. (Whatever value h is given at 0, no function has $f' = h$ at every point: h takes 0 and 1 but not $\frac{1}{2}$, so Darboux fails.)**

Marks: Darboux statement 1 · contradiction 2 · f_1 1 · f_2 1 · not constant 1.

Section C · Essays

10 marks each.

37 Bartle Thm 6.1.6

The Chain Rule via Carathéodory; why the naive proof fails; $\sin(x^2)$ and $|x^2-1|$.

KNOW FIRST

T06 The chain rule T05 Carathéodory's characterisation T02 Differentiable ⇒ continuous (and not conversely) M08 Composite bookkeeping

Theorem 6.1.6 (Chain Rule). Let I, J be intervals in $\mathbb{R}$, let $g : I \to \mathbb{R}$ and $f : J \to \mathbb{R}$ be functions such that $f(J) \subseteq I$, and let $c \in J$. If f is differentiable at c and if g is differentiable at $f(c)$, then the composite function $g \circ f$ is differentiable at c and $(g \circ f)'(c) = g'(f(c)) \cdot f'(c)$.

Why the naive proof fails. One writes $\frac{g(f(x)) - g(f(c))}{x - c} = \frac{g(f(x)) - g(f(c))}{f(x) - f(c)} \cdot \frac{f(x) - f(c)}{x - c}$, which divides by $f(x) - f(c)$. That can be 0 for $x \neq c$ arbitrarily near c: for $f(x) = x^2 \sin(1/x)$, $f(0) = 0$, $f(x) = 0$ at $x = \frac{1}{n\pi} \to 0$. Carathéodory's theorem removes the division.

SKELETON

(1) Carathéodory for f at c: $f(x) - f(c) = \varphi(x)(x - c)$. (2) Carathéodory for g at $d = f(c)$: $g(y) - g(d) = \psi(y)(y - d)$. (3) Substitute $y = f(x)$ and multiply out; the coefficient of $(x - c)$ is $\Phi(x) = \psi(f(x))\varphi(x)$? (continuous at c).

1. f differentiable at c: there is φ on J, continuous at c, with $f(x) - f(c) = \varphi(x)(x - c)$ for $x \in J$, and $\varphi(c) = f'(c)$.
   Carathéodory (⇒) for f.

2. With $d = f(c)$, g differentiable at d: there is ψ on I, continuous at d, with $g(y) - g(d) = \psi(y)(y - d)$ for $y \in I$, and $\psi(d) = g'(d)$.
   Carathéodory (⇒) for g.

3. For $x \in J$, put $y = f(x) \in I$: $g(f(x)) - g(f(c)) = \psi(f(x))\big(f(x) - f(c)\big) = \big[\psi(f(x))\,\varphi(x)\big](x - c)$.
   Substitute 2, then 1. Pure algebra, no division.

4. Let $\Phi(x) = \psi(f(x))\varphi(x)$. f is continuous at c (Thm 6.1.2) and ψ is continuous at $f(c)$, so $\psi \circ f$ is continuous at c; φ is continuous at c; hence Φ is continuous at c.
   Composite and product of continuous functions (Module 1).

5. $\Phi(c) = \psi(f(c))\varphi(c) = g'(f(c))\,f'(c)$.
   Values from steps 1 and 2.

6 By Carathéodory (⇐), $g \circ f$ is differentiable at c and $(g \circ f)'(c) = \Phi(c) = g'(f(c))f'(c)$. ∎
$g \circ f(x) - g \circ f(c) = \Phi(x)(x - c)$ with Φ continuous at c.

Applications.

$\sin(x^2)$. Inner $f(x) = x^2$, $f' = 2x$; outer $g = \sin$, $g'(y) = \cos y$ evaluated at $y = x^2$.

ANSWER $\dfrac{d}{dx}\sin(x^2) = 2x\cos(x^2)$. **Check x = 1:** $2\cos 1 = 1.0806$.

$|x^2 - 1|$. Inner $f(x) = x^2 - 1$, $f' = 2x$; outer $g(y) = |y|$, differentiable for $y \neq 0$ with $g'(y) = \operatorname{sgn} y$. So for $x \neq \pm 1$:

ANSWER $\dfrac{d}{dx}|x^2 - 1| = 2x\operatorname{sgn}(x^2 - 1)$: $= 2x$ **for** $|x| > 1$, $= -2x$ **for** $|x| < 1$.

At $x = \pm 1$ the outer function $|y|$ is not differentiable at $y = f(\pm 1) = 0$, so the rule is not available; directly, $f(\pm 1) = 0$ and $f'(\pm 1) = \pm 2 \neq 0$, so $|f|$ is not differentiable at ± 1 (one-sided quotients $+2$ and -2 at $x = 1$). Checks: x = 2: $|x^2 - 1|' = 4$; x = 0.5: $|f| = 1 - x^2$ gives -1 ✓.

Marks: statement 2 · naive failure 1 · Carathéodory for f and g 2 · identity 2 · continuity of Φ and conclusion 2 · examples 1.

38 Bartle Thm 6.2.4

The Mean Value Theorem; geometric meaning; hypotheses cannot be dropped.

KNOW FIRST

T12 The Mean Value Theorem T11 Rolle's Theorem P07 The big theorems of Module 1

Theorem 6.2.4 (Mean Value Theorem). Suppose that f is continuous on a closed interval $I = [a, b]$, and that f has a derivative in the open interval (a, b). Then there exists at least one point c in (a, b) such that $f(b) - f(a) = f'(c)(b - a)$.

SKELETON

Subtract the chord from f to get a function that is 0 at both ends, then use Rolle: $\varphi(x) = f(x) - (\text{chord line})$; $\varphi(a) = \varphi(b) = 0$; $\varphi'(c) = 0$

?

1. Define $\varphi(x) = f(x) - f(a) - \dfrac{f(b) - f(a)}{b - a}(x - a)$ on $[a, b]$.
   f minus the chord through $(a, f(a))$ and $(b, f(b))$.
2. φ is continuous on $[a, b]$ and differentiable on (a, b).
   f is, and the subtracted term is a linear function.
3. $\varphi(a) = 0$ and $\varphi(b) = f(b) - f(a) - \big(f(b) - f(a)\big) = 0$.
   Direct substitution.

4 By Rolle's Theorem there is $c \in (a, b)$ with $\varphi'(c) = 0$.
All Rolle hypotheses checked in 2 and 3.

5 $\varphi'(c) = f'(c) - \dfrac{f(b) - f(a)}{b - a} = 0$, so $f(b) - f(a) = f'(c)(b - a)$. ∎
Differentiate φ and rearrange.

Geometric meaning. $\dfrac{f(b) - f(a)}{b - a}$ is the slope of the chord; some point of the graph between a and b has a tangent parallel to the chord. Example: $f(x) = x^3$ on $[0, 2]$: chord slope 4, $3c^2 = 4$, $c = \dfrac{2}{\sqrt{3}} \approx 1.155 \in (0, 2)$.

Differentiability on (a, b) cannot be dropped. $f(x) = |x|$ on $[-1, 1]$ is continuous; chord slope $\dfrac{1 - 1}{2} = 0$, but $f'(x) = \pm 1 \neq 0$ wherever it exists, and $f'(0)$ does not exist. No c exists.

Continuity on $[a, b]$ cannot be dropped. $f(x) = x$ for $0 \leq x < 1$, $f(1) = 0$: f is differentiable on $(0, 1)$ with $f' = 1$, but is discontinuous at the endpoint 1. Chord slope $\dfrac{0 - 0}{1 - 0} = 0 \neq 1 = f'(c)$ for every c.

Consequence used later: if $f' = 0$ on an interval, f is constant; if $f' \geq 0$, f is increasing; if $|f'| \leq K$, f is Lipschitz.

Marks: statement 2 · φ and its properties 2 · Rolle and conclusion 2 · geometric meaning 1 · two counterexamples 2 · check 1.

39 Bartle Thm 6.2.12

Darboux's Theorem, both orderings of $f'(a), k, f'(b)$.

KNOW FIRST

T23 Darboux's Theorem T22 The local sign lemma T10 Relative extrema; the Interior Extremum Theorem P07 The big theorems of Module 1

Theorem 6.2.12 (Darboux). If f is differentiable on $I = [a, b]$ and if k is a number between $f'(a)$ and $f'(b)$, then there is at least one point c in (a, b) such that $f'(c) = k$.

Lemma 6.2.11 (used). Let $f : I \to \mathbb{R}$ have a derivative at $c \in I$. (a) If $f'(c) > 0$, there is $\delta > 0$ with $f(x) > f(c)$ for $x \in I$, $c < x < c + \delta$. (b) If $f'(c) < 0$, there is $\delta > 0$ with $f(x) > f(c)$ for $x \in I$, $c - \delta < x < c$.

SKELETON

Turn "$f' = k$" into "$g' = 0$" with $g(x) = kx - f(x)$; a maximum of g on $[a, b]$ exists, and it cannot sit at an endpoint because of the signs of $g'(a), g'(b)$. Then it is an interior extremum and $g' = 0$?

Case 1: $f'(a) < k < f'(b)$.

1. Let $g(x) = kx - f(x)$ on $[a, b]$. Then g is differentiable, $g'(x) = k - f'(x)$.
   $g'(c) = 0 \iff f'(c) = k.$

2. g is continuous on $[a, b]$, so it attains a maximum at some $c \in [a, b]$.
   Max–Min Theorem 5.3.4 (g continuous because differentiable).

3. $g'(a) = k - f'(a) > 0$. By Lemma 6.2.11(a) there are x just right of a with $g(x) > g(a)$, so the maximum is not at a.
   An endpoint max would need $g(x) \le g(a)$ nearby.

4. $g'(b) = k - f'(b) < 0$. By Lemma 6.2.11(b) there are x just left of b with $g(x) > g(b)$, so the maximum is not at b.
   Same argument at b.

5. Hence $c \in (a, b)$: an interior relative maximum, and $g'(c)$ exists, so $g'(c) = 0$ and $f'(c) = k$.
   Interior Extremum Theorem 6.2.1.

Case 2: $f'(a) > k > f'(b)$. Put $F = -f$ and $K = -k$. Then $F'(a) = -f'(a) < -k = K < -f'(b) = F'(b)$, which is Case 1. So there is $c \in (a, b)$ with $F'(c) = K$, i.e. $f'(c) = k$. ∎

Illustration. $f(x) = x^2$ on $[0, 2]$, $k = 1$: $f'(0) = 0 < 1 < 4 = f'(2)$. $g(x) = x - x^2$ has $g(0) = 0$, $g(2) = -2$, maximum $g(\frac{1}{2}) = \frac{1}{4}$ at $c = \frac{1}{2}$, and indeed $f'(\frac{1}{2}) = 1 = k$. For Case 2 use $-x^2$: $f'(0) = 0 > -1 > -4 = f'(2)$, again $c = \frac{1}{2}$.

Marks: statement 2 · g and maximum 2 · endpoints excluded via Lemma 3 · interior ⇒ g′ = 0 2 · Case 2 1.

40 Bartle Thm 6.2.7, Ex 6.2.10, 6.2.13

Monotonicity criterion; strict version; one-point positive derivative; $x - \sin x$.

KNOW FIRST

T14 Increasing and decreasing: the sign of f′ T12 The Mean Value Theorem T09 $x^2 \sin(1/x)$: differentiable, but f′ is not continuous P04 Limits of functions (Real Analysis I)

Theorem 6.2.7. Let $f : I \to \mathbb{R}$ be differentiable on the interval I. (a) f is increasing on I iff $f'(x) \geq 0$ for all $x \in I$. (b) f is decreasing on I iff $f'(x) \leq 0$ for all $x \in I$. (Increasing: $x_1 < x_2$ implies $f(x_1) \leq f(x_2)$.)

SKELETON

(⇐) take $x_1 < x_2$ and write $f(x_2) - f(x_1)$ with the MVT; sign of $f'(\xi)(x_2 - x_1)$? . (⇒) the difference quotient is ≥ 0; keep it in the limit.

(⇐) $f' \geq 0$ on I implies increasing. Let $x_1 < x_2$ in I. f is differentiable, hence continuous, on $[x_1, x_2]$. MVT: $f(x_2) - f(x_1) = f'(\xi)(x_2 - x_1)$ with $\xi \in (x_1, x_2)$. Since $f'(\xi) \geq 0$ and $x_2 - x_1 > 0$, $f(x_2) \geq f(x_1)$.

(⇒) increasing implies $f' \geq 0$. Let $c \in I$. For $x \in I$, $x \neq c$: $x > c$ gives $f(x) \geq f(c)$ and $x - c > 0$; $x < c$ gives $f(x) \leq f(c)$ and $x - c < 0$. In both cases $\dfrac{f(x) - f(c)}{x - c} \geq 0$. Letting $x \to c$, the order theorem for limits (4.2.6) gives $f'(c) \geq 0$. ∎ (b) follows by applying (a) to $-f$.

Strict version. If $f' > 0$ on I, then in the MVT line above $f'(\xi)(x_2 - x_1) > 0$, so $f(x_1) < f(x_2)$: strictly increasing. **Converse false:** $f(x) = x^3$ is strictly increasing but $f'(0) = 0$.

One point is not enough. $g(x) = x + 2x^2 \sin(1/x)$, $g(0) = 0$. $g'(0) = \lim \left(1 + 2x \sin \frac{1}{x}\right) = 1 > 0$ (squeeze). For $x \neq 0$: $g'(x) = 1 + 4x \sin \frac{1}{x} - 2 \cos \frac{1}{x}$. At $x_n = \frac{1}{2n\pi}$: $g' = 1 + 0 - 2 = -1 < 0$; at $y_n = \frac{1}{(2n+1)\pi}$: $g' = 1 + 0 + 2 = 3 > 0$. Both sequences tend to 0, so every neighbourhood of 0 contains points where $g' < 0$. By (⇒), g is not increasing on any neighbourhood of 0, although $g'(0) = 1 > 0$.

Application: $f(x) = x - \sin x$ on $\mathbb{R}$. $f'(x) = 1 - \cos x \geq 0$ for all x. By (a), f is increasing.

Strictly. Suppose $f(x_1) = f(x_2)$ for some $x_1 < x_2$. Increasing forces f constant on $[x_1, x_2]$, so $f' = 0$ on that whole interval; but $1 - \cos x = 0$ only at $x = 2k\pi$ (isolated points). Contradiction.

ANSWER $x - \sin x$ is strictly increasing on $\mathbb{R}$. Check: f(0) = 0, f(1) = 0.158, f(2) = 1.091, f(4) = 4.757.

Marks: statement 1 · (⇐) 2 · (⇒) 2 · strict and x^3 1 · single point example 2 · application 2.